BIO 280 PATHOPHYSIOLOGY · JOYCE UNIVERSITY

Dr. Garg's Exam 6 Master Study Guide

Modules 11 & 12 — Renal Disorders & the Respiratory System

ⓘ What Exam 6 covers

This unit is about **the kidneys and the lungs**. **Module 11 (Renal Disorders)** covers kidney function & labs, UTIs, nephrotic vs nephritic syndrome, acute kidney injury, chronic kidney disease, and other disorders (stones, PKD, bladder cancer, incontinence). **Module 12 (Respiratory)** covers gas exchange & V/Q mismatch, COPD, asthma and other chronic lung disease, the acute infections (pneumonia, flu, COVID), the emergencies (ARDS, PE, atelectasis), and chest trauma. Exam format: **50 questions (multiple choice + select-all-that-apply)**, proctored with Respondus LockDown Browser.

ⓘ How this guide works

- **Click any topic card** in the navigator to jump to that section. Click a title bar to expand or close it.
- **All answers are hidden** — try first, then click "Show answer." Practice includes **multiple choice** and **select-all-that-apply (SATA)**, like the real exam.
- **Each topic has:** clear explanation, a mnemonic, a "common confusion" warning, practice questions, and a clinical scenario.
- **Color code: amber = Module 11 (Renal)**, **teal = Module 12 (Respiratory)**.
- **To save as PDF:** use the button above, or press *Ctrl+P* / *Cmd+P* → "Save as PDF." All sections expand for printing.

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TOPIC 1

Kidney Function, Labs & UTI

MUST KNOW ▼

The kidney filters blood ~50 times a day and controls fluid, electrolytes, and pH. Start with the labs that measure it and the infections that attack it.

Kidney function tests

Test	What it means
GFR	Blood filtered per minute — the key measure of kidney function. **GFR falls** as disease worsens (end-stage < 15).
Creatinine	Muscle-breakdown waste cleared by the kidney. **As creatinine ↑, GFR ↓** — used to stage kidney disease.
BUN	Blood urea nitrogen (protein-breakdown waste); rises with declining function and dehydration.

Urinary tract infection: lower vs upper

Lower UTI	Upper UTI
Bladder/urethra — example: **acute cystitis**. Urgency, frequency, dysuria, suprapubic discomfort.	**Kidney** — example: **acute pyelonephritis**. Flank pain, fever, chills, systemic illness. Usually **ascending** from a lower UTI.

Memory hooks

Creatinine UP = GFR DOWN (they seesaw). **Lower UTI = cystitis** (burning/ frequency). **Upper UTI = pyelonephritis** (flank pain + fever).

⚠ Common confusion

A rising creatinine signals a **falling GFR**, not better kidneys. A lower UTI that ascends becomes **pyelonephritis** with fever and flank pain.

Practice Questions

1. A patient's creatinine is rising. What is happening to GFR, and why?

Show answer

GFR is falling. Creatinine is cleared by filtration, so when filtration drops it accumulates — the two move in opposite directions.

2. A patient has flank pain, fever, and chills. Lower or upper UTI, and which condition?

   Show answer

   Upper UTI → acute pyelonephritis. Flank pain + fever distinguish it from lower-tract cystitis.

3. **SATA:** Which are features of cystitis? (a) frequency, (b) dysuria, (c) high fever with flank pain, (d) urgency, (e) suprapubic discomfort.

   Show answer

   a, b, d, e. (c) points to upper-tract pyelonephritis.

Clinical scenario

A woman treated for a bladder infection now has a 102°F fever, chills, and right flank pain. **What happened?**

*Her **lower UTI ascended** to the kidney → **acute pyelonephritis**. New fever and flank pain signal kidney involvement needing prompt systemic treatment.*

TOPIC 2

□Nephrotic vs Nephritic Syndrome

MUST KNOW ▼

Two glomerular syndromes students mix up every term. The difference is **what leaks out of the glomerulus**.

	Nephrotic syndrome	Nephritic syndrome (AGN)
Patho	Membrane leaky **selectively to PROTEIN**	**Inflammation** (immune complexes) → leaky to **everything** (RBCs, WBCs, protein)
Main causes	**Diabetes mellitus, lupus**	**Untreated strep** (post-streptococcal GN)
Hallmarks	**Proteinuria >3.5 g/day (foamy urine), hypoalbuminemia, edema (periorbital/ extremities), hyperlipidemia**	**Hematuria ("cola-colored" urine)**, RBC/WBC in urine, **hypertension** (Na/fluid retention)

□Memory hooks

NephrOtic = prOtein & edema (foamy urine, low albumin; diabetes/lupus). **NephrItic = Inflammation & hematuria** (cola urine + hypertension; post-strep).

⚠ Common confusion

Nephrotic leaks **protein** (edema); **nephritic** is **inflammatory** and leaks **blood** (cola urine, HTN). Only nephritic is the classic post-strep picture.

□Practice Questions

1. A patient has proteinuria >3.5 g/day, foamy urine, low albumin, and periorbital edema. Which syndrome?

   Show answer

   Nephrotic syndrome. The glomerulus is selectively leaky to protein → heavy proteinuria, hypoalbuminemia, edema, hyperlipidemia (diabetes/lupus).

2. A child has cola-colored urine and hypertension two weeks after untreated strep throat. Which syndrome and mechanism?

Show answer

Nephritic syndrome (acute glomerulonephritis). Post-strep antigen–antibody complexes inflame the glomerulus → hematuria and Na/fluid retention (HTN).

3. Which finding best distinguishes nephritic from nephrotic syndrome?

Show answer

Hematuria with inflammation = nephritic; **heavy proteinuria with edema** = nephrotic.

Clinical scenario

A long-standing diabetic develops leg/facial swelling, foamy urine, and high cholesterol. **Which syndrome and why?**

Nephrotic syndrome *— chronic diabetes damaged the glomerular membrane, leaking protein → hypoalbuminemia (edema), foamy urine, and compensatory hyperlipidemia.*

TOPIC 3

⚡ Acute Kidney Injury (AKI)

MUST KNOW ▼

Acute kidney injury (AKI) = sudden decline in kidney function (hours–days), often **reversible** if caught early. Classify by **where** the problem is.

Type	Problem	Causes
Pre-renal	↓ **blood flow TO the kidney** (nephrons initially healthy)	Hypovolemia (dehydration, hemorrhage, shock), heart failure, atherosclerosis
Intra-renal	**Damage INSIDE the kidney**	**Acute tubular necrosis**, glomerulonephritis, **nephrotoxins** (NSAIDs, IV contrast, vancomycin)
Post-renal	**Obstruction AFTER the kidney** → back-pressure	**BPH**, tumor, stones → hydroureter/ hydronephrosis

Acute tubular necrosis (ATN)

The most common intra-renal cause: tubular cells die (ischemia or toxins) → slough off → **tubular obstruction** → oliguria and edema.

Phases of recovery

Phase	What happens
1. Oliguric	**<400 mL/24 hr**; ↑ BUN/creatinine, fatigue, fluid-overload risk
2. Diuretic	**2–4 L/day**; risk of **dehydration, hypovolemia, hypotension**; BUN/Cr stabilize
3. Recovery	Urine normalizes (1–2 L/day); GFR improves; BUN/Cr decrease

Memory hooks

Pre = flow problem, Intra = internal damage, Post = plumbing/obstruction.

Recovery: **Oliguric** → **Diuretic** → **Recovery**.

⚠ Common confusion

The **diuretic phase** is deceptive — the big urine output isn't "all better"; tubules can't concentrate yet, so watch for **dehydration and electrolyte loss**.

Practice Questions

1. A patient in hypovolemic shock develops AKI. Which classification, and why are nephrons initially undamaged?

   Show answer

   Pre-renal AKI. Reduced blood flow *to* the kidney — nephrons are fine but can't filter what they don't receive. Restoring perfusion often reverses it.

2. **SATA:** Which cause intra-renal AKI? (a) acute tubular necrosis, (b) NSAIDs/IV contrast, (c) BPH obstruction, (d) glomerulonephritis, (e) dehydration.

   Show answer

   a, b, d. BPH is **post-renal**; dehydration is **pre-renal**.

3. In which AKI phase is the patient at highest risk for dehydration and hypotension?

   Show answer

   The **diuretic phase** — output surges (2–4 L/day) but the kidney can't concentrate it yet.

Clinical scenario

An older man can't urinate, has a distended bladder, rising creatinine, and an enlarged prostate. **Which AKI type?**

Post-renal AKI *from* ***BPH*** *obstructing outflow → urine backs up (hydroureter → hydronephrosis), pressure rises, GFR falls. Relieving the obstruction is the fix.*

TOPIC 4

Chronic Kidney Disease (CKD)

MUST KNOW ▼

Chronic kidney disease (CKD) = permanent, progressive nephron loss → declining GFR → fluid, waste, and electrolyte buildup. It's a **multisystem** disorder.

Most common causes: diabetes and hypertension.

Increased (↑)	Decreased (↓)
Creatinine & BUN, **potassium**, **hydrogen ions (acidosis)**, phosphate	**GFR**, **calcium**, **vitamin D**, **erythropoietin (→ anemia)**

Why those labs

- ↓ **EPO** → **anemia** (fatigue, pallor).
- ↑ **potassium (hyperkalemia)** — can't excrete it → dangerous for the heart.
- ↑ **H^+** → **metabolic acidosis.**
- ↓ **vitamin D activation** → ↓ **calcium** → bone disease; phosphate rises.
- Fluid retention → edema, hypertension.

Management: low sodium/potassium diet, avoid magnesium-containing laxatives, vitamin D + calcium supplements; end-stage (GFR < 15) → **dialysis or transplant**.

Memory hooks

CKD's top causes = **Diabetes & Hypertension.** Labs: **everything up EXCEPT calcium, GFR, vitamin D, and EPO. ↓EPO → anemia; can't dump K^+ → hyperkalemia.**

⚠ Common confusion

In CKD **calcium goes DOWN** (impaired vitamin D activation) while phosphate goes up — don't lump calcium with the "everything is high" group.

Practice Questions

1. What are the two most common causes of CKD?

Show answer

Diabetes and hypertension — both damage the kidney's small vessels over time.

2. A CKD patient has fatigue and pallor. Which lab change explains it?

   Show answer

   Decreased erythropoietin → fewer RBCs → anemia.

3. **SATA:** In CKD, which labs DECREASE? (a) GFR, (b) calcium, (c) potassium, (d) vitamin D, (e) erythropoietin.

   Show answer

   a, b, d, e. Potassium rises (can't be excreted).

Clinical scenario

A patient with 20 years of poorly controlled diabetes and hypertension has GFR 12, high potassium, low calcium, and anemia. **What stage, and what's next?**

End-stage CKD. *Nephron loss dropped GFR below 15, causing hyperkalemia, low calcium (low vitamin D), and anemia (low EPO). Typically requires* ***dialysis or transplant****.*

TOPIC 5

□Stones, PKD, Bladder Cancer & Incontinence

HIGH YIELD

Common urologic disorders that show up in practice and on the exam.

Disorder	Key points
Nephrolithiasis (kidney stones)	Concentrated minerals crystallize (most often **calcium**). Risk: **hypercalcemia (hyperparathyroidism), high uric acid, low fluid intake**. **Abrupt severe flank pain radiating to the groin**, vomiting, ± hematuria — pain begins when the stone moves from kidney into ureter. Prevention: **hydration**.
Polycystic kidney disease	**Inherited** — multiple cysts enlarge kidneys and destroy tissue. Often asymptomatic early; then flank pain, **hematuria, nocturia, hypertension**, ± palpable kidneys.
Bladder cancer	Most commonly **urothelial carcinoma**. Classic sign: **painless hematuria**; may mimic UTI; risk with smoking/chemicals.
Urinary incontinence	**Overflow** = retention/obstruction (BPH; males > females; infection risk). **Stress** = sphincter weakness (leak with cough/laugh; females, pregnancies, age). **Urge/OAB** = overactive detrusor (sudden strong urges).

□Memory hooks

Stones = colicky flank-to-groin pain (hydrate to prevent). **PKD = inherited cysts → big kidneys + HTN. Bladder cancer = PAINLESS hematuria.** Incontinence: Overflow / Stress / Urge.

⚠ Common confusion

Hematuria clues: stones = **painful**; bladder cancer = **painless**; nephritic syndrome = cola-colored with inflammation.

□Practice Questions

1. A patient has sudden severe flank pain radiating to the groin with vomiting. Which disorder, and when does pain start?

Show answer

Nephrolithiasis. Pain begins as the stone **passes from kidney into the ureter**; hydration prevents stones by keeping urine dilute.

2. A patient reports painless blood in the urine and a smoking history. What must be ruled out?

   Show answer

   Bladder cancer. Painless hematuria (often urothelial carcinoma) is the classic presentation, especially with smoking.

3. A man with BPH has chronic bladder distension and dribbling. Which incontinence type?

   Show answer

   Overflow incontinence — obstruction/retention makes the overfull bladder leak; stasis also raises infection risk.

Clinical scenario

A 35-year-old with a family history of kidney disease has hypertension, flank discomfort, nocturia, and bilaterally enlarged lumpy kidneys. **Diagnosis?**

Polycystic kidney disease *— inherited cysts enlarge the kidneys and replace functional tissue, producing hypertension, hematuria/nocturia, and palpable kidneys.*

TOPIC 6

Gas Exchange, V/Q & Lung Sounds

MUST KNOW

Respiration is a **three-step sequence**. Every lung disease breaks one of these steps — name the broken step and the symptoms follow.

Component	What it is	Diseases that break it
Ventilation	Moving air in/out of alveoli (**obstructive** problems)	Asthma, COPD, OSA, cystic fibrosis, bronchitis
Diffusion	O_2/CO_2 crossing the alveolar–capillary membrane	Pulmonary edema, pneumonia, ARDS, emphysema
Perfusion	Blood flow to the alveoli	**Pulmonary embolism**, pulmonary hypertension

V/Q mismatch

- **Shunt** = **perfusion to an area that isn't ventilated** (blood but no air) → COPD, mucus plugging, pneumonia.
- **Dead space** = **ventilation to an area that isn't perfused** (air but no blood) → classic in **pulmonary embolism**.

Hypoxia vs hypoxemia, and lung sounds

- **Hypoxemia** = low oxygen in the **blood**; **hypoxia** = low oxygen in the **tissues**. Signs: **tachycardia, cyanosis, restlessness, dyspnea**.
- **Crackles (rales)** = air through **fluid**. **Wheezes** = **narrowed airways**. **Rhonchi** = **mucus** in large airways (snoring). **Stridor** = upper-airway narrowing.

Memory hooks

Shunt = blood, no air; Dead space = air, no blood (PE). Crackles = fluid, Wheeze = tight airway, Rhonchi = mucus.

⚠ Common confusion

Hypoxemia (blood) precedes **hypoxia** (tissues). A **PE** is a **perfusion** problem — ventilation is fine, blood can't reach that region.

Practice Questions

1. A PE blocks blood flow to a ventilated lung region. Which V/Q abnormality?

   Show answer

   > **Dead space** — ventilation without perfusion. PE is fundamentally a **perfusion** problem.

2. You hear crackles. What do they indicate?

   Show answer

   > **Fluid in the alveoli** (pneumonia, pulmonary edema, heart failure). Wheezes = tight airways; rhonchi = mucus.

3. Difference between hypoxemia and hypoxia?

   Show answer

   > **Hypoxemia** = low oxygen in the **blood**; **hypoxia** = low oxygen in the **tissues**.

Clinical scenario

A patient breathes rapidly but oxygen stays low; a large clot sits in a pulmonary artery. **Why is oxygen low despite good breathing?**

*The clot creates **dead space** — that region is ventilated but not perfused, so no gas exchange occurs. The broken step is **perfusion**.*

TOPIC 7

□COPD: Chronic Bronchitis vs Emphysema

MUST KNOW

COPD is progressive, not-fully-reversible airflow obstruction — an umbrella for **chronic bronchitis** and **emphysema** (often both). The core problem is getting air **out**.

	Chronic bronchitis	Emphysema
Patho	**Hypersecretion of mucus** + airway inflammation from a chronic **irritant/toxin** (not infection)	**Destruction of alveoli + loss of surfactant/elastic recoil** → can't exhale fully (air trapping)
Signs	**Chronic productive cough**, dyspnea, hypoxemia, **hypercapnia**	**Barrel chest, prolonged expiration**, dyspnea, pursed-lip breathing

The hypoxic drive

In chronic COPD, CO_2 is always high, so the body relies on **low oxygen** to stimulate breathing (**hypoxic drive**). **Too much oxygen can slow their breathing**, letting CO_2 climb dangerously — titrate O_2 to a target rather than high-flow.

□Memory hooks

Chronic Bronchitis = mucus & cough (hypercapnia). **Emphysema = barrel chest & air trapping.** COPD = "getting air **out** is the problem." **Too much O_2 suppresses the hypoxic drive.**

⚠ Common confusion

Chronic bronchitis is from an **irritant/toxin** (smoking), **not** infection. COPD obstruction is about **exhalation**, unlike restrictive disease (trouble inhaling).

□Practice Questions

1. A long-time smoker has a barrel chest, prolonged expiration, and destroyed alveoli. Which COPD component?

   Show answer

   Emphysema — alveolar destruction with loss of recoil traps air.

2. Why give oxygen cautiously in chronic COPD?

Show answer

They rely on a **hypoxic drive**; too much oxygen slows respirations and lets CO_2 build up. Titrate to a target SpO_2.

3. What causes chronic bronchitis, and its hallmark signs?

Show answer

A chronic **irritant/toxin** (smoking), not infection → mucus hypersecretion → **chronic productive cough**, dyspnea, hypoxemia, hypercapnia.

Clinical scenario

A COPD patient with SpO_2 86% is placed on high-flow oxygen and becomes drowsy with slowed breathing. **What happened?**

*Excess oxygen blunted the **hypoxic drive**, so respirations slowed and CO_2 rose (CO_2 narcosis → drowsiness). Oxygen should be titrated to a controlled target.*

TOPIC 8

□Asthma & Other Chronic Lung Disease

MUST KNOW ▼

The other chronic lung diseases: asthma (reversible), plus occupational, sleep, vascular, and genetic conditions.

Disease	Patho & signs
Asthma	Two problems: **bronchoconstriction** + **inflammation** (mucus, swelling). Reversible. S/S: **dyspnea/chest tightness, wheezing, cough (worse at night)**; triggers: exercise, allergens
Cystic fibrosis	**Autosomal recessive** → abnormally thick mucus → **recurrent lung infections, hypoxia, poor growth**
Pneumoconiosis ("black lung")	Inhaled dust (**coal, silica, asbestos**) → scarring → obstruction and poor lung expansion
Obstructive sleep apnea	Throat muscles relax and **block the airway** in sleep. S/S: **daytime fatigue, memory issues**; risk: obesity, male, alcohol. Can cause **pulmonary HTN → right HF**
Pulmonary hypertension	High pulmonary artery pressure → **right ventricular strain → right-sided heart failure**. Secondary causes: COPD, PE, OSA, fibrosis

□Memory hooks

Asthma = bronchoconstriction + inflammation (reversible). **CF = autosomal recessive, thick mucus. Pneumoconiosis = dust. OSA/chronic lung disease → pulmonary HTN → right heart failure.**

⚠ Common confusion

Asthma vs COPD: asthma is **reversible** and trigger-driven; COPD is **chronic, not fully reversible** (usually smoking).

□Practice Questions

1. What are the two core mechanisms of an asthma attack?

Show answer

> **Bronchoconstriction** plus **inflammation** (mucus and swelling) → obstruction, wheezing, cough.

2. Cystic fibrosis inheritance pattern and core lung problem?

   Show answer

   > **Autosomal recessive**; abnormally **thick mucus** → recurrent infections, hypoxia, poor growth.

3. How does OSA lead to right-sided heart failure?

   Show answer

   > Repeated nighttime hypoxia raises pulmonary pressure → **pulmonary hypertension** → right-ventricular strain → **right-sided heart failure**.

Clinical scenario

A retired coal miner has progressive dyspnea with lung scarring; a young patient has episodic wheezing and night cough triggered by cats. **Name each.**

*The miner has **pneumoconiosis** from coal dust → fibrosis. The younger patient has **asthma** — reversible bronchoconstriction and inflammation from an allergen.*

TOPIC 9

Acute Infections: Pneumonia, Flu & COVID

MUST KNOW

The acute respiratory infections — from a mild chest cold to lobar pneumonia and the two big viruses.

Upper respiratory infections (usually viral, self-limiting)

Pharyngitis (sore throat), **sinusitis** (facial pain/congestion), **rhinitis** (runny nose), **laryngitis** (hoarse voice). Treatment is usually **supportive care**.

Condition	Patho & signs
Acute bronchitis	Acute, usually **viral** ("chest cold"); inflamed bronchi, cough, often **rhonchi**; self-limiting — usually **no antibiotics**
Pneumonia	Acute, usually **bacterial**, **localized** to one lobe; **alveoli fill with fluid/ exudate**. S/S: **fever, productive cough, crackles, hypoxia, dyspnea, confusion** (elderly)
Influenza	Virus attacks respiratory **epithelial cells**. S/S: **high fever, body aches, headache, dry cough, sore throat**
COVID-19	Targets **alveolar** epithelial cells (type II pneumocytes) via the **ACE-2 receptor** → deep lower-respiratory inflammation and rapid oxygen decline

Memory hooks

Acute bronchitis = viral chest cold (rhonchi, no antibiotics). **Pneumonia = bacterial, localized, fluid-filled alveoli. Flu = upper epithelium; COVID = alveoli via ACE-2.**

⚠ Common confusion

Acute vs chronic bronchitis: acute = viral, self-limiting; chronic = long-term irritant (smoking), part of COPD.

Practice Questions

1. A patient has fever, productive cough, crackles in one lobe, and hypoxia. Which infection, and what fills the alveoli?

Show answer

> **Pneumonia** (usually bacterial, localized). Alveoli fill with **fluid/inflammatory exudate**, impairing gas exchange.

2. Why does COVID drop oxygen more than influenza?

   Show answer

   > COVID targets **alveolar** cells via the **ACE-2 receptor** — right where gas exchange happens. Flu mainly affects upper-airway epithelium.

3. Should acute viral bronchitis get antibiotics?

   Show answer

   > **No.** It's usually viral and self-limiting — supportive care is the treatment.

Clinical scenario

An 80-year-old is brought in confused with fever, wet cough, crackles over the right lower lobe, and SpO_2 88%. **Diagnosis?**

Pneumonia *— a localized alveolar infection filling air sacs with fluid. In the elderly,* ***new confusion*** *is often an early sign.*

TOPIC 10

Emergencies: ARDS, PE & Atelectasis

MUST KNOW

The acute lung **emergencies** — fast onset, fast deterioration.

Condition	Patho & signs
ARDS	**Massive pulmonary inflammation → leaky capillaries → alveolar edema**. Triggers: **SEPSIS, burns, trauma**. S/S: severe dyspnea, rapid breathing, refractory hypoxemia
Pulmonary embolism	A **dislodged thrombus** (usually leg DVT) blocks a pulmonary artery → dead space. Risk: **DVT, immobility, femur fracture, birth control, smoking, obesity**. S/S: **sudden dyspnea, pleuritic chest pain, hemoptysis, hypoxemia**
Atelectasis	**Collapse of alveoli** (bronchiole blockage or **post-op** shallow breathing) → that area can't exchange gas

Memory hooks

ARDS = body-wide inflammatory hit (sepsis/burns/trauma) → leaky alveoli. **PE = sudden dyspnea + pleuritic pain + hemoptysis. Atelectasis = collapsed alveoli** (post-op — incentive spirometry).

⚠ Common confusion

ARDS vs pneumonia: pneumonia is a **localized infection**; ARDS is **diffuse whole-lung inflammation** from a systemic trigger with refractory hypoxemia.

Practice Questions

1. A septic patient develops severe dyspnea and hypoxemia that barely improves with oxygen, with diffuse lung involvement. Which condition?

   Show answer

   ARDS. Widespread inflammation makes alveolar capillaries leak, flooding alveoli → diffuse, refractory hypoxemia.

2. **SATA:** Which raise PE risk? (a) DVT, (b) prolonged immobility, (c) daily walking, (d) recent femur fracture, (e) smoking/oral contraceptives.

Show answer

a, b, d, e. Walking (c) *reduces* risk.

3. Why is a post-op patient at risk for atelectasis, and what prevents it?

Show answer

Pain and shallow breathing let alveoli collapse. **Deep breathing, coughing, and incentive spirometry** keep them open.

Clinical scenario

Three days after knee surgery, a patient becomes suddenly short of breath with sharp chest pain, coughs blood, SpO_2 85%. **What emergency, and its origin?**

Pulmonary embolism *— a clot from a leg DVT (post-op immobility) lodged in a pulmonary artery. Blocked perfusion creates dead space → sudden dyspnea, pleuritic pain, hemoptysis, hypoxemia.*

TOPIC 11

Chest Trauma & Pneumothorax

HIGH YIELD

Chest trauma disrupts **ventilation** (and sometimes circulation). Know the mechanics of each.

Injury	Patho & signs
Flail chest	**Multiple ribs broken in multiple places** → loose segment moves **paradoxically** (inward on inspiration) → decreased ventilation
Pneumothorax	**Air in the pleural space** separates visceral & parietal pleura → the lung **collapses**. **Hemothorax** = blood in the pleural space
Tension pneumothorax	**Trapped air keeps building** → shifts the mediastinum and **compresses the heart/great vessels**. **MEDICAL EMERGENCY.** S/S: **hypotension, tachycardia, tracheal deviation AWAY from the affected side, JVD, absent breath sounds on one side**

Memory hooks

Flail chest = paradoxical movement (sucks IN on inspiration). **Pneumothorax = air in pleura → collapse. Tension = compresses the heart** → hypotension + **trachea AWAY** + JVD — decompress now.

⚠ Common confusion

A simple **pneumothorax** is a stable collapsed lung. A **tension** pneumothorax adds **hemodynamic collapse** — look for hypotension, JVD, and tracheal deviation away.

Practice Questions

1. After a crash, part of the chest wall moves inward during inspiration. Which injury, and what is this called?

   Show answer

   Flail chest — multiple ribs broken in multiple places create a free segment with **paradoxical movement**, reducing ventilation.

2. A trauma patient has hypotension, absent left breath sounds, JVD, and the trachea deviated right. Which emergency?

Show answer

Tension pneumothorax (left side). Trapped air compresses the heart/great vessels and pushes the trachea **away** — needs immediate decompression.

3. What is a hemothorax?

Show answer

Blood in the pleural space (vs air in a pneumothorax) — it also compresses the lung and impairs ventilation.

Clinical scenario

A stabbing victim becomes rapidly hypotensive and tachycardic with distended neck veins, no right breath sounds, and the trachea shifted left. **What is happening and why is it lethal?**

Tension pneumothorax. *Air enters the pleural space and can't escape; the building pressure collapses the lung and* ***compresses the heart/great vessels*** *so blood can't return — obstructive shock. Requires immediate decompression.*

Exam Cheat Sheet — Night-Before Review

Module 11 — Renal Disorders

- **Labs: creatinine UP = GFR DOWN**; BUN rises with declining function. End-stage GFR < 15.
- **UTI:** lower = **cystitis** (bladder; frequency/dysuria); upper = **pyelonephritis** (kidney; flank pain + fever).
- **NephrOtic** = prOtein leak → proteinuria >3.5 g, hypoalbuminemia, edema, hyperlipidemia (diabetes/lupus). **NephrItic** = Inflammation → hematuria (cola urine), HTN (post-strep).
- **AKI:** Pre-renal (↓flow — dehydration/shock), Intra-renal (ATN, nephrotoxins, GN), Post-renal (obstruction — BPH). Phases: **Oliguric → Diuretic → Recovery**.
- **CKD:** #1 causes **diabetes & hypertension**. UP: Cr/BUN, K^+, H^+ (acidosis), phosphate. DOWN: **GFR, calcium, vitamin D, EPO (→ anemia)**.

- **Stones** = colicky flank→groin pain (hydrate to prevent). **PKD** = inherited cysts + HTN. **Bladder cancer** = **painless hematuria**. Incontinence: overflow / stress / urge.

Module 12 — Respiratory

- **3 steps:** Ventilation (obstructive), Diffusion (membrane/fluid), Perfusion (blood flow). **Shunt = blood no air; Dead space = air no blood (PE).**
- **Lung sounds:** crackles = fluid; wheeze = tight airways; rhonchi = mucus; stridor = upper airway.
- **COPD:** chronic bronchitis (mucus, cough, hypercapnia) + emphysema (barrel chest, air trapping). **Hypoxic drive** — don't over-oxygenate.
- **Asthma** = bronchoconstriction + inflammation (reversible). **CF** = autosomal recessive thick mucus. **Pneumoconiosis** = dust. **OSA/chronic lung dz → pulmonary HTN → right HF.**
- **Pneumonia** = bacterial, localized, fluid-filled alveoli. **Acute bronchitis** = viral, self-limiting. **Flu** = upper epithelium; **COVID** = alveoli via ACE-2.
- **ARDS** = diffuse inflammation (sepsis/burns/trauma) → leaky alveoli, refractory hypoxemia. **PE** = clot → dead space. **Atelectasis** = collapsed alveoli (post-op).
- **Trauma:** flail chest = paradoxical movement; pneumothorax = air in pleura; **tension pneumothorax** = hypotension + JVD + **trachea deviates AWAY** (emergency).

□Don't-Mix-These-Up

Easily confused	The key difference
Cystitis vs pyelonephritis	Lower UTI (bladder; frequency/dysuria) vs upper UTI (kidney; flank pain + fever).
Nephrotic vs nephritic	Protein leak + edema (diabetes/lupus) vs inflammation + hematuria/HTN (post-strep).
Pre/intra/post-renal AKI	Flow to kidney / damage in kidney / obstruction after kidney.
AKI vs CKD	Sudden & often reversible vs permanent, progressive nephron loss.
Painful vs painless hematuria	Stones (colicky pain) vs bladder cancer (painless).
Shunt vs dead space	Perfusion without ventilation vs ventilation without perfusion (PE).

Chronic bronchitis vs emphysema	Mucus/cough/hypercapnia vs alveolar destruction/barrel chest.
Acute vs chronic bronchitis	Viral, self-limiting vs long-term irritant (smoking), part of COPD.
Asthma vs COPD	Reversible bronchospasm/inflammation vs chronic, not-fully-reversible obstruction.
Crackles vs wheezes vs rhonchi	Fluid vs tight airways vs mucus.
Pneumothorax vs tension pneumothorax	Stable collapsed lung vs hemodynamic collapse (trachea away, JVD, hypotension).

□Glossary

GFR — glomerular filtration rate; the main measure of kidney function (falls with disease).

Creatinine / BUN — waste markers that rise as kidney function declines.

Cystitis / Pyelonephritis — bladder infection / kidney infection.

Nephrotic syndrome — protein-leaking glomerulus → proteinuria, hypoalbuminemia, edema, hyperlipidemia.

Nephritic syndrome — inflamed glomerulus → hematuria, hypertension (often post-strep).

Acute tubular necrosis — the most common intra-renal AKI; tubular cell death.

Hydronephrosis — kidney swelling from obstructed urine backing up.

CKD — chronic kidney disease; progressive nephron loss (diabetes, hypertension).

Nephrolithiasis — kidney stones.

Ventilation / Diffusion / Perfusion — the three steps of respiration.

V/Q mismatch — imbalance of ventilation and perfusion (shunt vs dead space).

Hypoxemia / Hypoxia — low oxygen in the blood / in the tissues.

COPD — chronic obstructive pulmonary disease (chronic bronchitis + emphysema).

Hypoxic drive — reliance on low O_2 to stimulate breathing in chronic COPD.

Pneumonia — localized alveolar infection filling air sacs with fluid.

ARDS — diffuse alveolar inflammation/edema → refractory hypoxemia.

Pulmonary embolism — a clot blocking a pulmonary artery (dead space).

Atelectasis — alveolar collapse (often post-op).

Tension pneumothorax — trapped pleural air compressing the heart — an emergency.

Final Study Tips

- **Renal: watch GFR & creatinine** — they move in opposite directions. Know pre/intra/post-renal AKI and the CKD lab pattern cold.
- **Lungs: categorize by the broken step** — ventilation, diffusion, or perfusion. That alone predicts most symptoms.
- **Lock in the opposite pairs:** nephrotic vs nephritic, cystitis vs pyelonephritis, chronic bronchitis vs emphysema, shunt vs dead space, crackles vs wheezes vs rhonchi.
- **Know the emergencies:** tension pneumothorax, pulmonary embolism, ARDS, and obstructive AKI.
- **Creatinine/BUN** = kidney-function markers (rise as GFR falls). **Lung sounds**: crackles = fluid, wheezes = tight airways, rhonchi = mucus.
- **Practice SATA carefully** — read each option as its own true/false statement; partial credit usually isn't given.
- **Use the cheat sheet** the night before AND the morning of the exam, then self-test with the answers hidden. Format: 50 questions, Respondus LockDown Browser — prep your space ahead of time.

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You've got this. Master each disorder's one hallmark, and the scenarios practically answer themselves.

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